

Courier Management System: Salesforce CRM Implementation

Project Description:

The Courier Management System (CMS) is a cloud-based Salesforce CRM application that brings courier activities such as customer registration, parcel booking, branch coordination, delivery assignment, status updates and billing into one environment. It stores courier information in connected custom objects, applies validation rules, automates shipment status updates with Salesforce Flow, controls access with profiles and roles, and presents operational information through reports and a dashboard. The project was submitted as part of the Naan Mudhalvan practical project documentation by a team of five students (Team ID SWTID-2026-1564).

User story: As a branch manager, I want to manage shipments, agents and delivery updates from one system so that daily courier operations can be monitored without depending on multiple manual records.

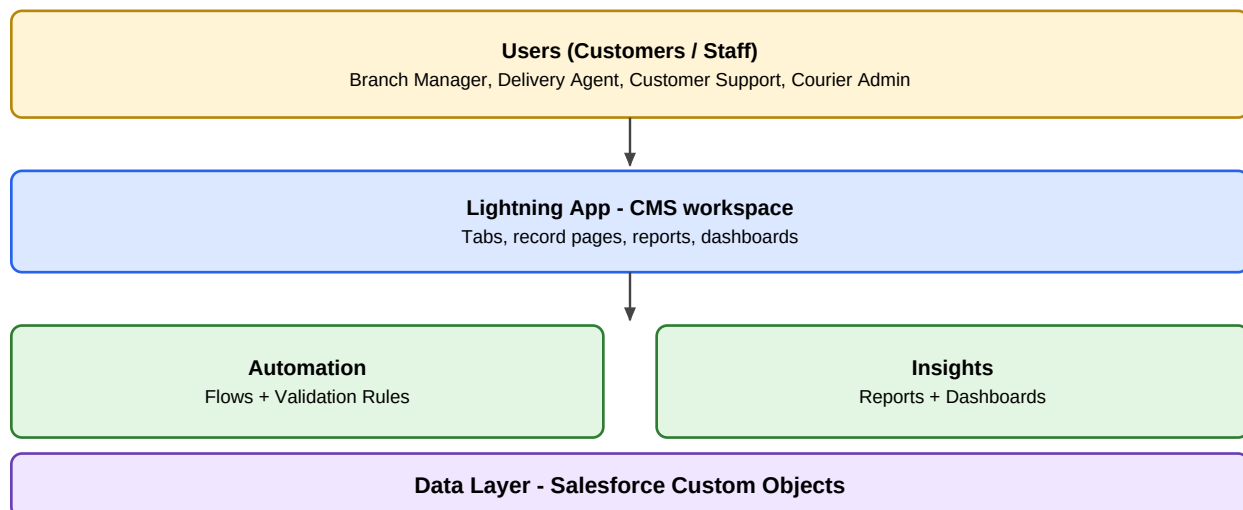
Scenarios

Scenario 1: A branch manager records a new customer and creates a shipment with source, destination, weight, price and expected date. The system auto-generates a tracking number, and validation rules reject a zero weight or a past expected date.

Scenario 2: A delivery agent logs a Delivery Update with status Delayed. Validation requires a remark, and the record-triggered Flow copies the status to the related shipment, so support staff see the latest status immediately.

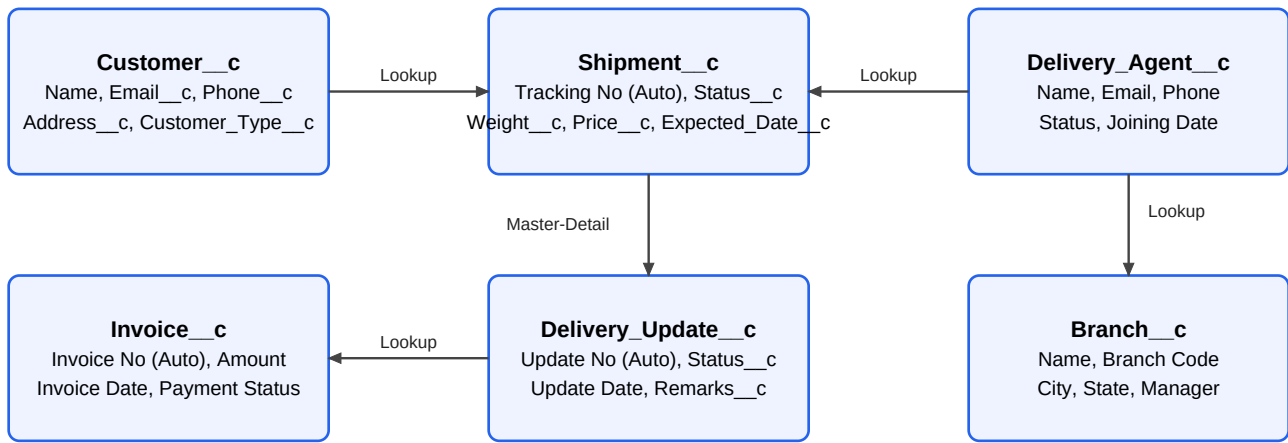
Scenario 3: A branch manager opens the Courier Management Dashboard to review the shipment status donut chart, agent performance bar chart and revenue-by-branch column chart.

Technical Architecture:



Description: Users work in the Lightning App; Flows and validation rules enforce data quality and automate status updates; reports and dashboards summarize the stored data; all records live in custom Salesforce objects.

Data model (ER diagram):



Customer 1-to-many Shipment | Agent 1-to-many Shipment | Branch 1-to-many Agent | Shipment 1-to-many Update / Invoice

Pre-requisites:

- Salesforce Developer Org (free account) and Salesforce Setup familiarity.
- Understanding of objects, fields, relationships and record pages.
- Basic knowledge of validation rule formulas and Flow Builder.
- Knowledge of profiles, roles, users, reports and dashboards.

Project Workflow:

- **Milestone 1:** Ideation and requirement analysis
- **Milestone 2:** Salesforce developer account and platform setup
- **Milestone 3:** Data model - objects, tabs, fields and relationships
- **Milestone 4:** Validation rules, page layouts and Flow automation
- **Milestone 5:** Lightning app, profiles, roles and users
- **Milestone 6:** Reports and dashboard
- **Milestone 7:** Testing and conclusion

Milestone 1: Ideation and Requirement Analysis

Activity 1.1 - Problem statement. The branch manager, customer, delivery agent and support staff all struggle because tracking is spread across records, updates are not centralized and assignments are shared manually.

Activity 1.2 - Brainstorming. Four options were compared: manual register, spreadsheet, separate web portal and Salesforce CRM. Salesforce was selected because it is cloud based and provides automation, security and reports.

Idea	Strength	Limitation
Manual Register	Easy to start	Hard to search and update
Spreadsheet	Simple digital storage	Limited automation and coordination
Separate Web Portal	Can be customized	Requires additional infrastructure
Salesforce CRM	Cloud based with automation	Needs configuration and learning

Activity 1.3 - Requirements. Functional requirements FR-01 to FR-07: customer records; shipment records with source, destination, weight, price and status; shipment-to-agent association; delivery status updates; branch and invoice information; reports and dashboards; validation rules. Non-functional requirements cover usability, security, reliability, performance, availability and scalability.

Activity 1.4 - Planning. Six Agile-style sprints: Platform Setup (3 points), Data Modeling (5), Automation (5), Security (5), Reporting (4) and Dashboard (4).

Milestone 2: Salesforce Developer Account

A Salesforce developer account provides a free cloud environment for building the CMS. Typical setup:

- Open the Salesforce Developer sign-up page.
- Enter the requested personal and organization information.
- Activate the account through the verification email.
- Set the password and complete the initial login.
- Open **Setup** to begin configuring the application.

Milestone 3: Data Model

Activity 3.1 - Create custom objects. Go to Setup > Object Manager > Create > Custom Object. Enter the label, plural label and record name type, enable the required optional features and save.

Object	Purpose	Record Name
Customer__c	Customer profile and contact information	Customer Name (Text)
Shipment__c	Parcel and tracking information	Tracking Number (Auto Number)
Delivery_Agent__c	Delivery personnel details	Agent Name (Text)
Branch__c	Courier branch information	Branch Name (Text)

Object	Purpose	Record Name
Delivery_Update__c	Shipment status history	Update Number (Auto Number)
Invoice__c	Billing and payment information	Invoice Number (Auto Number)

Activity 3.2 - Create tabs. Create tabs for Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice so the objects are easy to reach from the Lightning app.

Activity 3.3 - Create fields. Use Fields & Relationships > New, choose the data type, set the label, required setting or picklist values and save.

Customer fields

Field	API Name	Type	Required
Customer Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Address	Address__c	Text Area	Yes
Customer Type	Customer_Type__c	Picklist	Yes
City	City__c	Text	No
State	State__c	Text	No
Pincode	Pincode__c	Text	No
Is Active	Is_Active__c	Checkbox	No

Shipment fields

Field	API Name	Type	Required
Tracking Number	Name	Auto Number	Yes
Customer	Customer__c	Lookup	Yes
Source Address	Source_Address__c	Text Area	Yes
Destination Address	Destination_Address__c	Text Area	Yes
Weight	Weight__c	Number	Yes
Price	Price__c	Currency	Yes
Status	Status__c	Picklist	Yes
Expected Date	Expected_Date__c	Date	Yes
Assigned Agent	Assigned_Agent__c	Lookup	No

Field	API Name	Type	Required
Shipment Type	Shipment_Type__c	Picklist	Yes

Other objects - important fields

Object	Important Fields
Delivery Agent	Agent Name, Email, Phone, Branch, Status, Joining Date
Branch	Branch Name, Branch Code, City, State, Manager Name
Delivery Update	Update Number, Shipment, Status, Update Date, Remarks
Invoice	Invoice Number, Shipment, Amount, Invoice Date, Payment Status

Activity 3.4 - Relationships.

Child	Field	Parent	Relationship
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery Agent	Lookup
Delivery Agent	Branch__c	Branch	Lookup
Delivery Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Milestone 4: Validation Rules, Page Layouts and Flow

Activity 4.1 - Validation rules. A validation rule shows an error and blocks saving when its condition evaluates to true.

Object	Rule	Formula / Condition	Message
Customer	Email required	ISBLANK(Email__c)	Email is required.
Customer	Phone length	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Customer	Pincode length	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Shipment	Positive weight	Weight__c <= 0	Weight must be greater than zero.
Shipment	Positive price	Price__c <= 0	Price must be greater than zero.
Shipment	Future expected date	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Delivery Agent	Email format check	NOT(CONTAINS(Email__c, "@"))	Enter a valid email address.

Object	Rule	Formula / Condition	Message
Branch	Branch code required	ISBLANK(Branch_Code__c)	Branch Code is mandatory.

Delayed shipment remark rule (Delivery Update):

```
AND(
  ISPICKVAL(Status__c, "Delayed"),
  ISBLANK(Remarks__c)
)
```

Error message: Remarks are required when status is Delayed.

Activity 4.2 - Page layouts. Group fields into sections so users enter information in a logical order.

Object	Suggested Sections
Customer	Customer Information; Address Details
Shipment	Shipment Details; Address Information; Delivery Information
Delivery Agent	Agent Details; Additional Information
Branch	Branch Information
Delivery Update	Update Information
Invoice	Invoice Details

Activity 4.3 - Record-triggered Flow. The Flow updates the shipment status whenever a Delivery Update is created: *Delivery Update Created > Find Related Shipment > Copy Status > Save Shipment*.

- Create a new Record-Triggered Flow on **Delivery_Update__c**; trigger: record is created.
- Use Update Records to find Shipment__c where Id equals the Delivery Update's Shipment__c.
- Set Shipment__c.Status__c to the Delivery Update Status__c.
- Save and activate the flow.

Milestone 5: Lightning App, Profiles, Roles and Users

Activity 5.1 - Lightning App. Create a Lightning App named **Courier Management System** with navigation items: Customers, Shipments, Delivery Agents, Branches, Delivery Updates, Invoices, Reports and Dashboards.

Activity 5.2 - Profiles.

Profile	Main Purpose
Courier Admin	Full configuration and administrative control
Branch Manager	Manage branch-level shipment and agent operations
Delivery Agent	Update assigned shipment delivery information
Customer Support	Handle customer and tracking enquiries

Activity 5.3 - Role hierarchy. Courier Admin > Branch Manager > Delivery Agent. Customer Support can be placed under the Branch Manager when a separate support role is required. The hierarchy organizes reporting relationships and data visibility.

Activity 5.4 - Users. Create one user per type with a unique username, valid email, role and profile (Courier Admin, Branch Manager, Delivery Agent, Customer Support). Available licenses depend on the org.

Milestone 6: Reports and Dashboard

Report	Main Fields / Grouping
Shipment Status Summary	Tracking Number, Customer, Status, Expected Date, Assigned Agent
Agent Performance	Agent Name, Branch, Shipment Count, Status
Revenue by Branch	Tracking Number, Customer, Price, Invoice Status, Branch, Month
Customer Summary	Customer details and related shipment information

Dashboard: *Courier Management Dashboard*

Component	Source	Visualization
Shipment Status Overview	Shipment Status Summary	Donut Chart
Agent Performance	Agent Performance Report	Bar Chart
Revenue by Branch	Revenue Report	Column Chart

Milestone 7: Testing and Conclusion

Test ID	Scenario	Expected Result
T01	Create a valid customer	Customer record saves successfully
T02	Enter an invalid phone length	Validation message appears
T03	Enter a non-six-digit pincode	Pincode validation prevents save
T04	Enter zero shipment weight	Weight validation prevents save
T05	Enter a past expected date	Date validation prevents save
T06	Create a Delivery Update	Related shipment status is updated by Flow
T07	Create report	Correct shipment data is displayed
T08	Open dashboard	Charts display the configured report data
T09	Login as restricted user	Only permitted objects/actions are available

Advantages: centralized storage; fewer repeated manual status updates; validation prevents selected data-entry errors; Flow synchronizes shipment status; reports and dashboards give visibility; profiles and roles control

access.

Limitations: depends on Salesforce configuration and availability; initial setup needs time and platform knowledge; SMS, GPS and mobile features need additional implementation; licenses and permissions must be configured correctly.

Conclusion:

The Courier Management System demonstrates how Salesforce can be configured to support courier operations through structured data, relationships, validation, automation, security and analytics. It connects customers, shipments, delivery agents, branches, updates and invoices into a common application and shows the complete development cycle from problem identification and requirements to design, configuration, testing and reporting.

Future scope: mobile-friendly shipment tracking; GPS-based delivery location integration; SMS or email notifications; additional branch-level analytics; AI-assisted route or workload optimization; customer self-service tracking. These are proposed extensions and are not claimed as implemented.

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